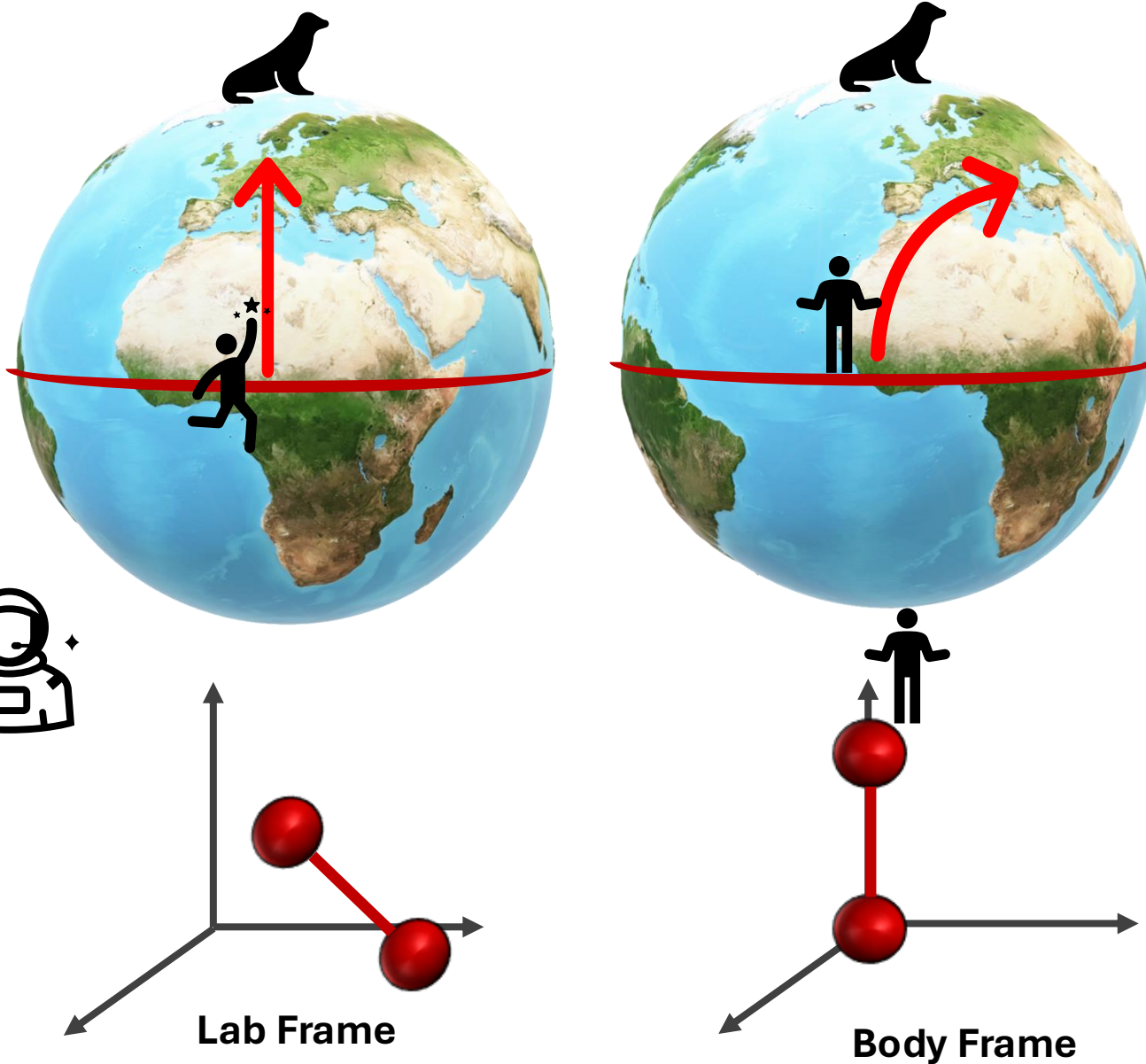


Electron-Nuclear Momentum Transfer Beyond Born-Oppenheimer: A Phase-Space Perspective

Mansi Bhati
Princeton University

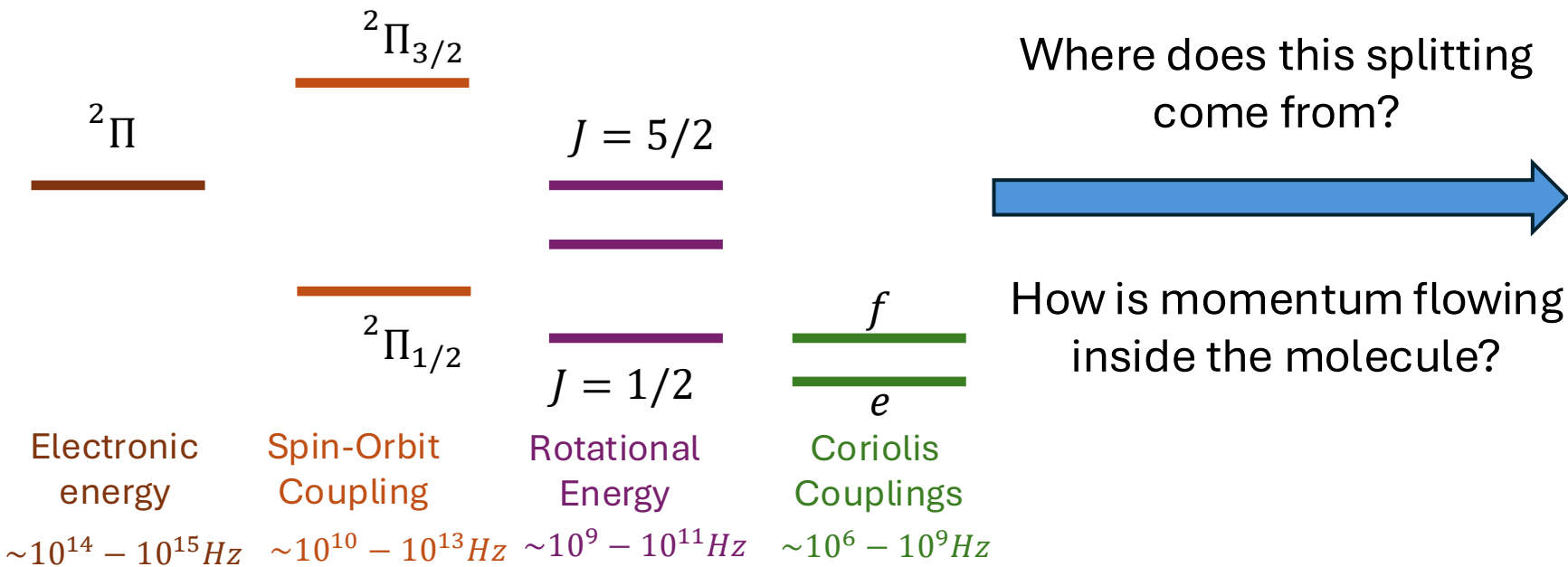
Subotnik Lab
Third-Year Seminar
April 3, 2026

Motion Depends on the Frame of Reference



- Inertial frame (**Space**): Trajectory is straight
- Rotating Frame (**Earth**): Trajectory appears curved
- This deflection is the **Coriolis effect**
- Changing frame can mix different components of motion

What Does Spectra Reveal About Momentum Transfer ?



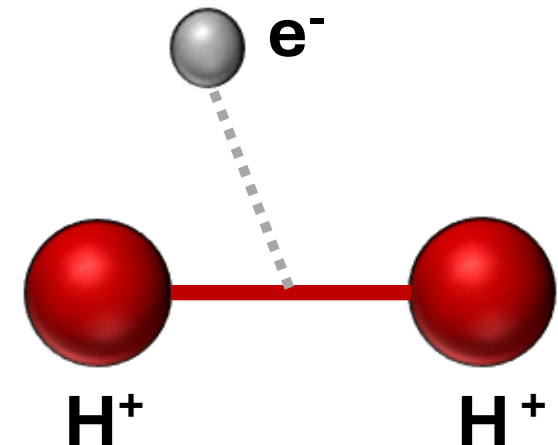
Where does this splitting come from?

How is momentum flowing inside the molecule?

Methods:

- **Exact Quantum** (benchmark)
- **Born-Oppenheimer** (standard)
- **Phase-Space** (our proposed approach)

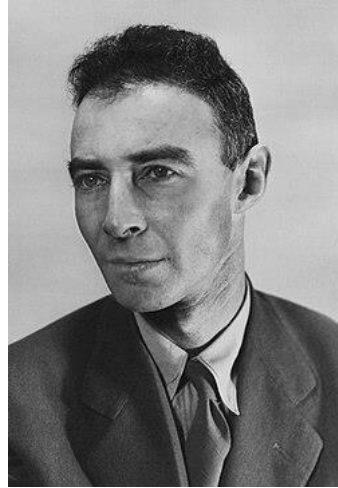
- Simplest molecule with **electron-nuclear coupling**
- No electron-electron correlation $\rightarrow$ clean test system



Born-Oppenheimer Hamiltonian



Max Born



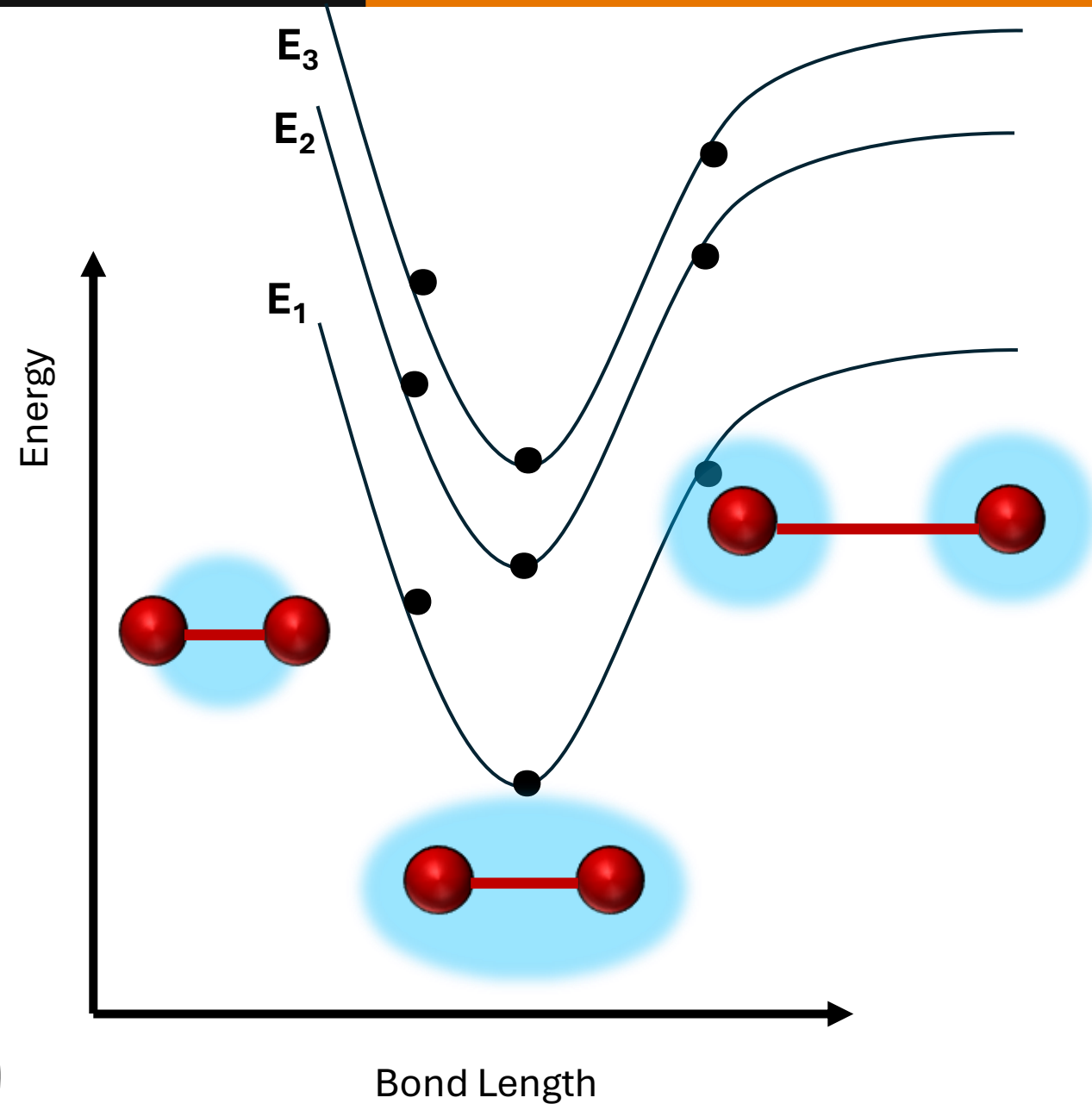
J. Robert Oppenheimer

Mass of nuclei (1836 a.u.) $\gg$ mass of electron (1 a.u.)

Nuclear motion is much slower than electrons

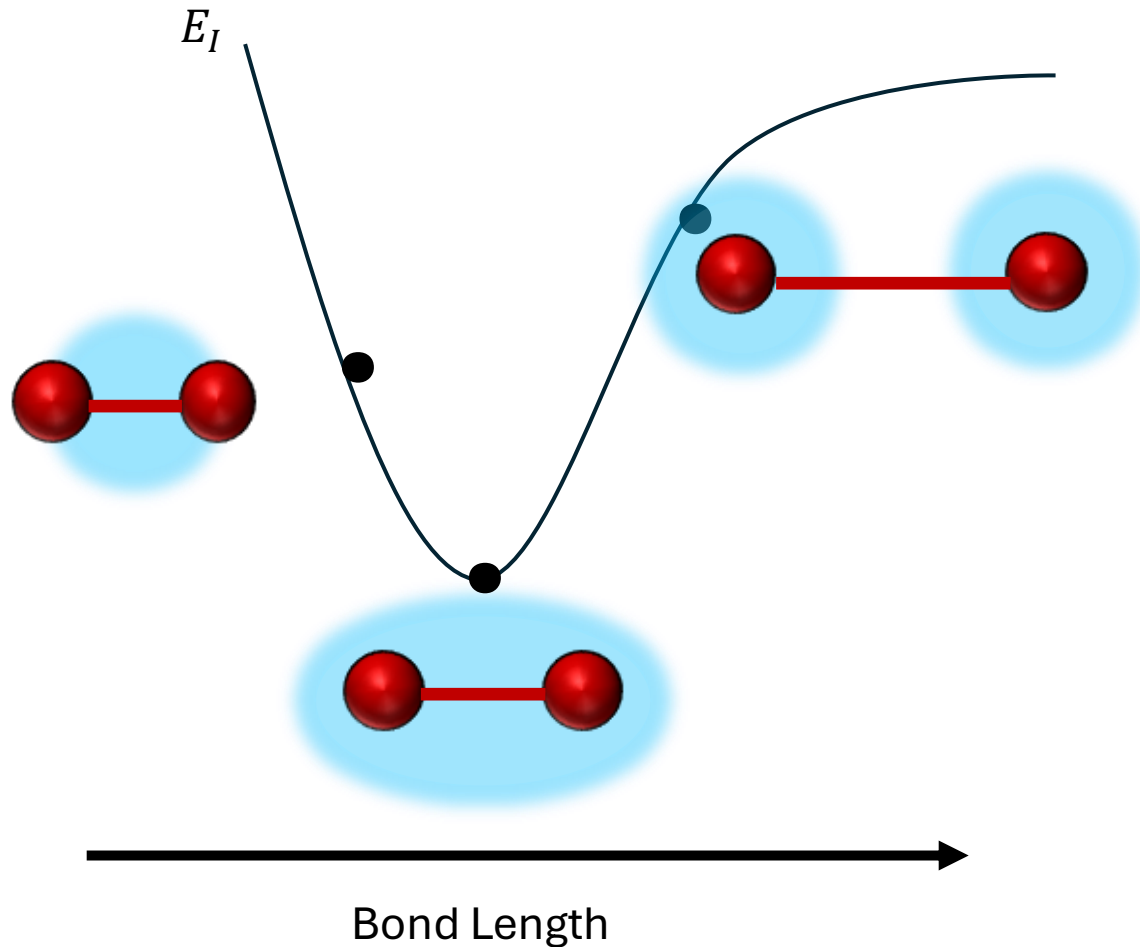
Electrons instantly adapt to change in nuclear position

$$\hat{H}(\mathbf{R}) = \frac{\mathbf{P}^2}{2M} + \hat{H}_{el}(\hat{\mathbf{r}}, \mathbf{R})$$



A Small but Deep Puzzle

Each of the energy surface is well separated from the other surfaces



Electrons are relaxed at each nuclear geometry.
They appear to carry no momentum!

Books $\equiv$ Electrons



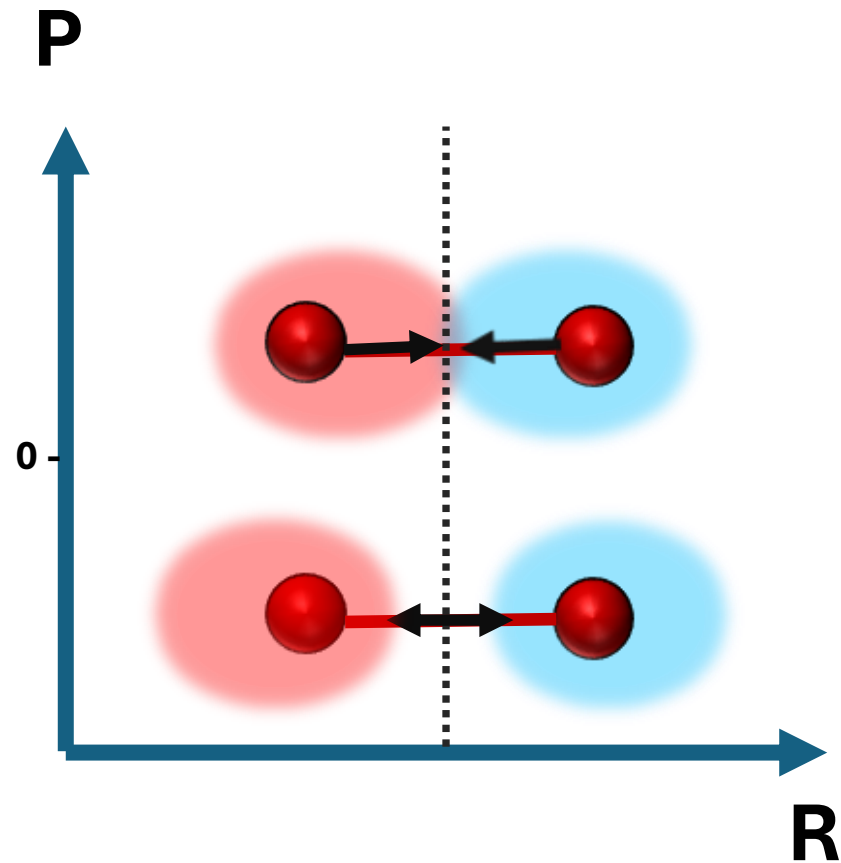
Me $\equiv$ Nuclei

Moving nuclei drags the electron cloud

$$\langle \psi_I(\mathbf{R}) | \hat{\mathbf{p}}_e | \psi_I(\mathbf{R}) \rangle = 0$$

Where did the electronic momentum go?

Phase Space treats position and momentum on equal footing



Electronic momentum should react to how nuclei move!

Phase-space accounts for how electrons respond to different nuclear motions not just positions

$$\langle \psi_I(\mathbf{R}, \mathbf{P}) | \hat{\mathbf{p}}_e | \psi_I(\mathbf{R}, \mathbf{P}) \rangle \neq 0$$

Hidden electronic momentum reappears!!

$$\hat{\mathbf{P}}_{\text{nuc}}^{\text{effective}}(\mathbf{R}) = \hat{\mathbf{P}}_{\text{nuc}}(\mathbf{R}) - \hbar \hat{\mathbf{A}}(\mathbf{R})$$

Berry connection

Derivative coupling

$$\hat{\mathbf{A}}(\mathbf{R}) = i \langle \psi_I(\mathbf{R}) | \nabla_{\mathbf{R}} | \psi_I(\mathbf{R}) \rangle$$

Our Proposed Phase-Space Electronic Hamiltonian

$$\hat{H}(\mathbf{R}, \hat{\mathbf{P}}) = \frac{(\hat{\mathbf{P}} - \hbar \hat{\mathbf{A}}(\mathbf{R}))^2}{2M} + \frac{\hat{\mathbf{p}}_e^2}{2m} + \hat{V}(\hat{\mathbf{r}}, \mathbf{R})$$

$\hat{\mathbf{A}}(\mathbf{R})$ terms are computationally demanding and can be gauge-dependent when there are spin degeneracies.

How do we make this physics practical for a simulation?

1. Make Nuclear Momentum $\mathbf{P}$ classical $\hat{\mathbf{P}} \rightarrow \mathbf{P}$
2. We designed one-electron $\hat{\mathbf{\Gamma}}(\mathbf{R})$ to replace $\hat{\mathbf{A}}(\mathbf{R})$

$$\hat{H}(\mathbf{R}, \mathbf{P}) = \frac{(\mathbf{P} - \hbar \hat{\mathbf{\Gamma}}(\mathbf{R}))^2}{2M} + \frac{\hat{\mathbf{p}}_e^2}{2m} + \hat{V}(\hat{\mathbf{r}}, \mathbf{R})$$

Designing the $\hat{\Gamma}$ Term

1. Electrons should follow the nuclear motion.

2. Translationally and Rotationally invariant vectors

$$\left(-i\hbar \sum_A \frac{\partial}{\partial \mathbf{R}_A} + \hat{\mathbf{p}}_e \right) |\psi_e(\mathbf{R}, \mathbf{P})\rangle = 0$$

Sum over all nuclei

Electronic momentum

Electron Translation Factor (ETF) $\left(-i\hbar \sum_A \hat{\mathbf{r}}_A + \hat{\mathbf{p}}_e \right) |\psi_e(\mathbf{R}, \mathbf{P})\rangle = 0$

Corrects for total linear momentum

Electron Rotational Factor (ERF) $\left(-i\hbar \sum_A \mathbf{R}_A \times \hat{\mathbf{r}}_A + \hat{\mathbf{l}}_e + \hat{\mathbf{s}}_e \right) |\psi_e(\mathbf{R}, \mathbf{P})\rangle = 0$

Corrects for total angular momentum



It recovers the electronic momentum

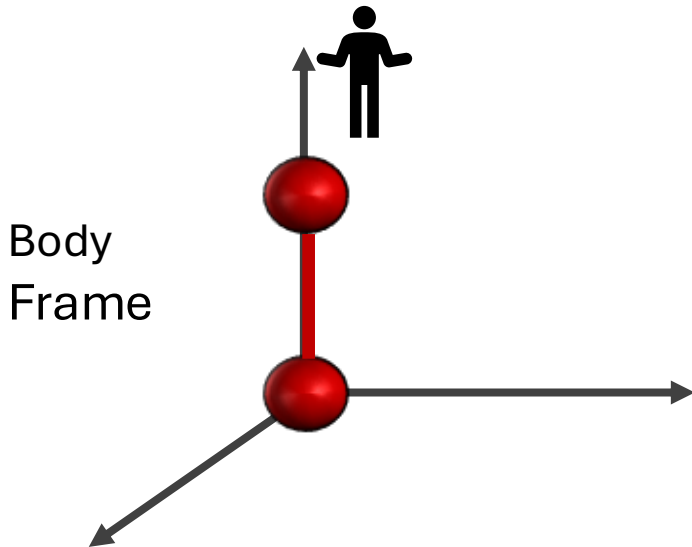
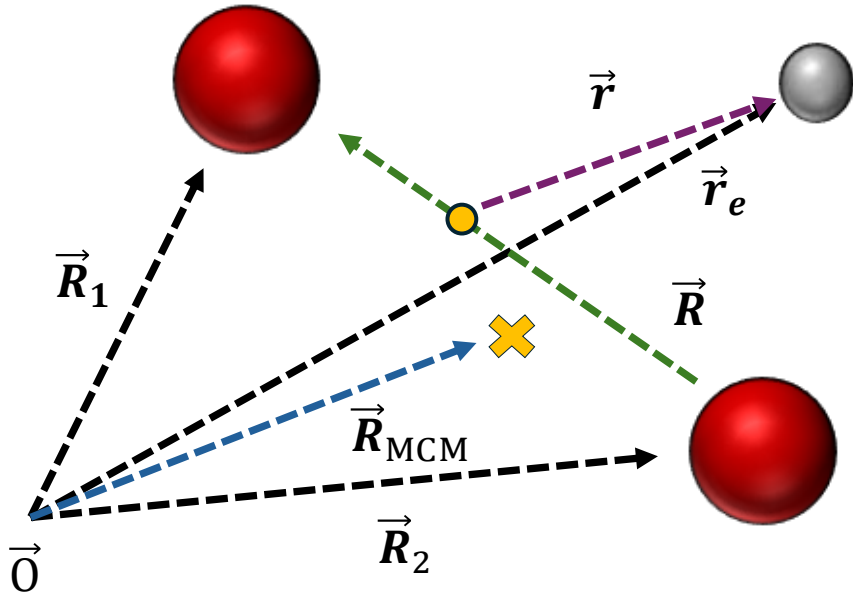


It handles the spin degeneracy cases



Its fast and scales well with molecular size

Separating Internal Motion from overall Rotation



We can solve the Hamiltonian numerically.
Grid size 100^9 consumes **8 x 10⁹ GB!** → Use symmetries

$$\hat{H}_{Jacobi} = \cancel{\frac{\hat{p}_M^2}{2\mu_{total}}} + \frac{\hat{P}_R^2}{2\mu_{12}} + \frac{\hat{p}_r^2}{2\mu_r} + V(\hat{R}, \hat{r})$$

Removing Translations: Separate center-of-mass motion
→ Reduces to 6 degrees of freedom

Removing Rotations

Body frame: **internuclear axis = z-axis**

→ separates:

- Internal motion (vibration + electronic)
- Overall rotations
- Rotation doesn't disappear → **becomes couplings**

Hamiltonian in the body frame

$$\hat{H}_{\text{exact}} = \frac{1}{2\mu} \left[-\frac{1}{R} \frac{\partial^2}{\partial R^2} R + \frac{\hat{N}^2}{R^2} - \frac{1}{r} \frac{\partial^2}{\partial r^2} r + \frac{\hat{L}^2}{r^2} \right] + V(r, R, \gamma)$$

$\hat{N}$ Nuclear Angular Momentum
 $\hat{L}$ Electronic Angular Momentum
 $\hat{J}$ Total Angular Momentum

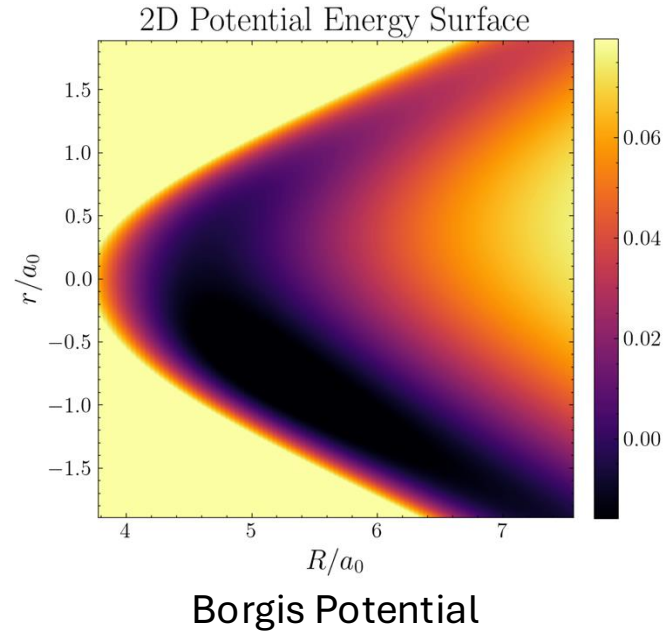
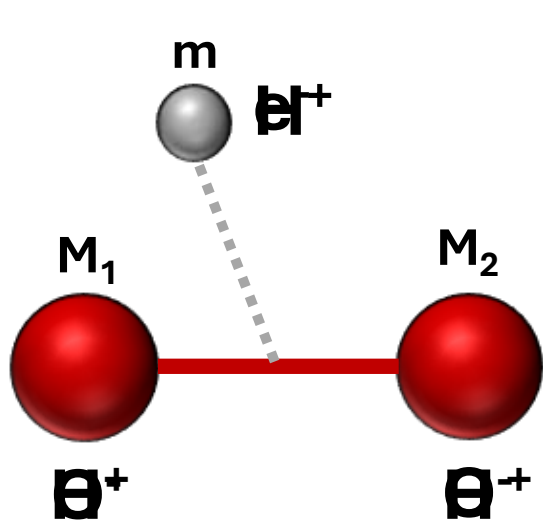
$$\frac{\hat{N}^2}{R^2} = \frac{(\hat{J} - \hat{L})^2}{R^2}$$

$$\frac{\hat{N}^2}{R^2} = \frac{\hat{J}^2 + \hat{L}^2 - 2\hat{J} \cdot \hat{L}}{R^2}$$

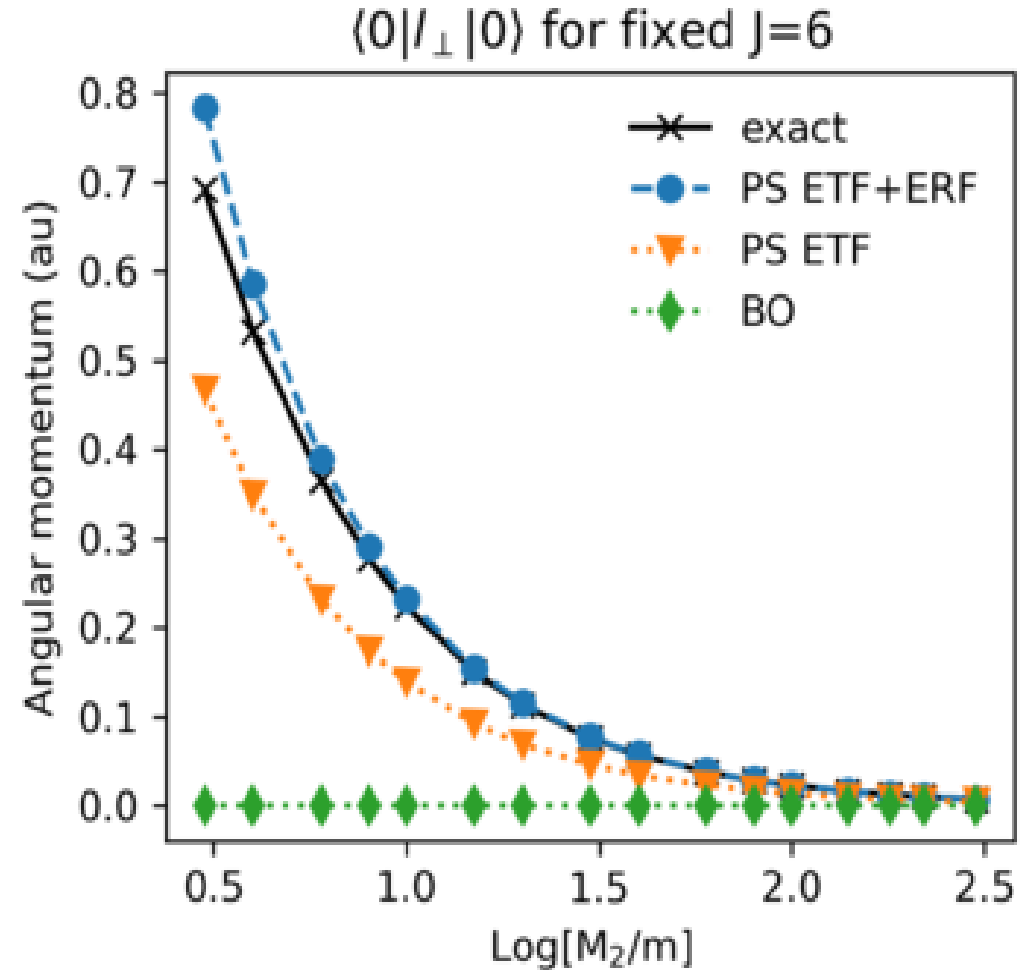
$\hat{J}^2$ Total Rotation of the whole molecule
 $\hat{L}^2$ Rotation of electrons around the nuclei
 $\hat{J} \cdot \hat{L}$ Quantum Coriolis Coupling

Born-Oppenheimer: $\hat{N}^2 \approx \hat{J}^2$
 Phase-Space Coupling: $\mathbf{P} \cdot \hat{\mathbf{\Gamma}} \simeq \langle 0, P_\theta \cdot \hat{l}_z, P_\phi \cdot \hat{l}_y \rangle$

Results: Comparing Electronic Angular Momentum



- Born-Oppenheimer has **net zero electronic momentum**.
- In Phase-Space, since the electrons move along with nuclei, so it imparts electronic momentum **at almost same accuracy** as exact calculations!

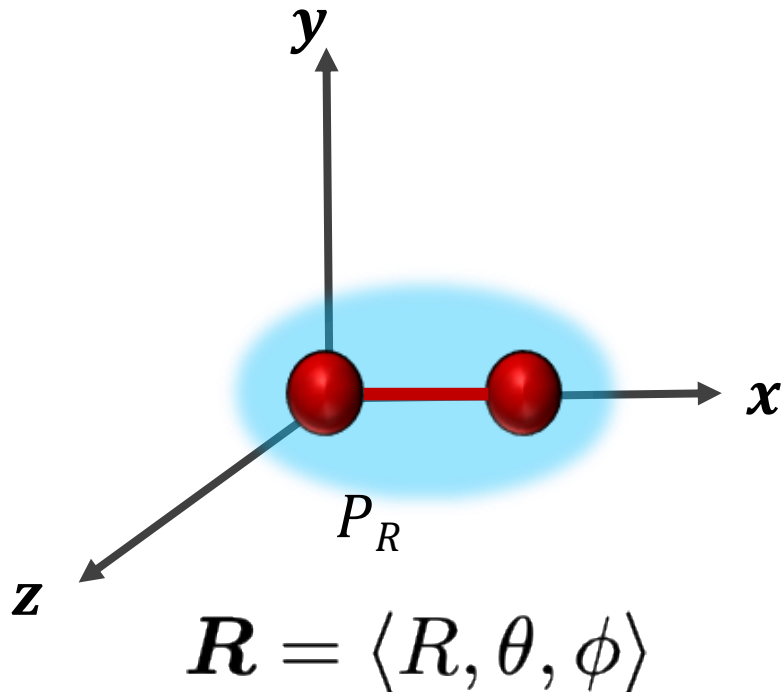


ETF: Electrons can translate with the nuclei

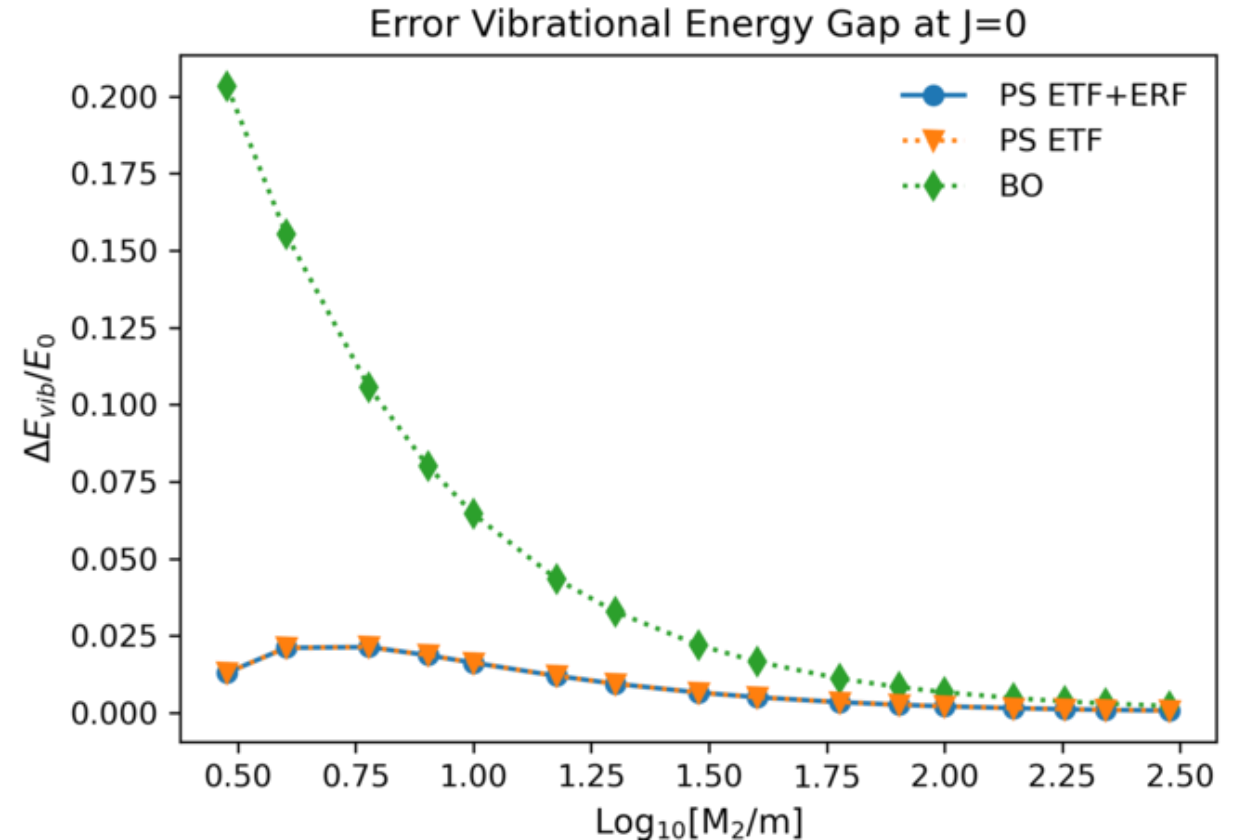
ERF: Electrons can rotate with the nuclei

Results: Comparing Vibrational Energies

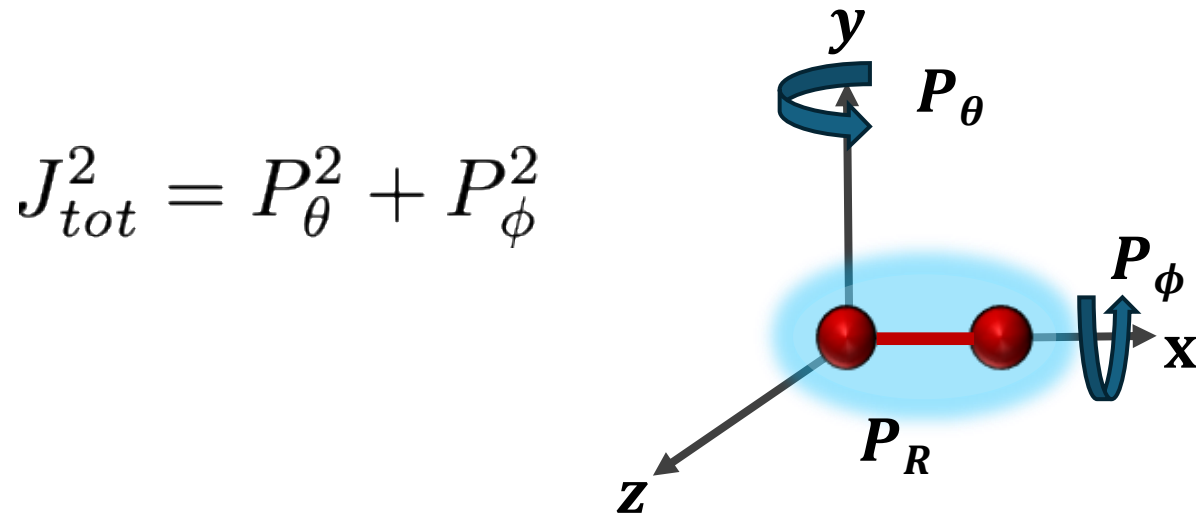
- Low mass ratio nuclei move faster so electron nuclear motion becomes strongly coupled.
- BO fails because electrons are essentially frozen
- Since electrons are vibrating with nuclei:
→ **ETF improves vibrational energies significantly**



ETF: Corrects Linear Momentum
ERF: Corrects Angular Momentum

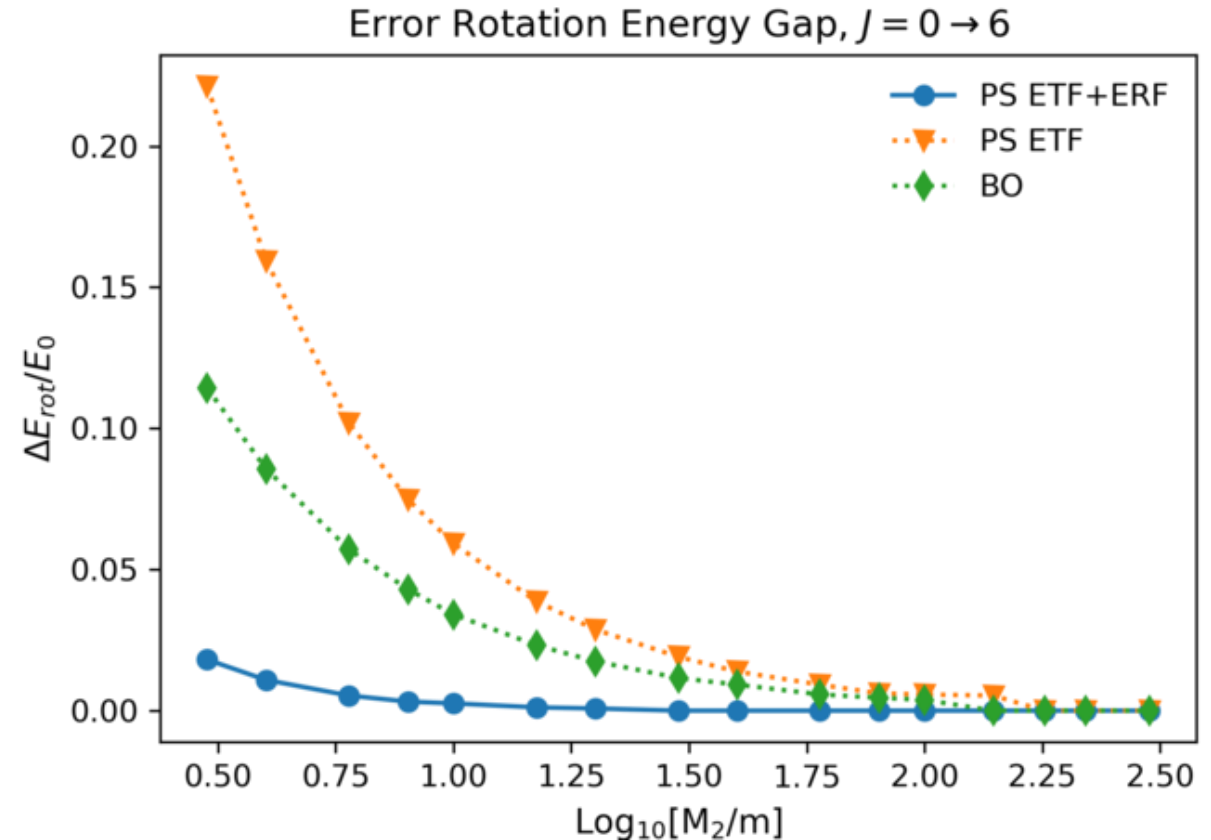


Results: Comparing Rotational Energies



- In exact calculations, angular momentum is shared between nuclei and electrons
→ **Quantum Coriolis effect**
- **ERF improves rotational energies significantly**
- The semiclassical rotations match well with full quantum rotation.

ETF : Electron Translation Factor
ERF : Electron Rotation Factor

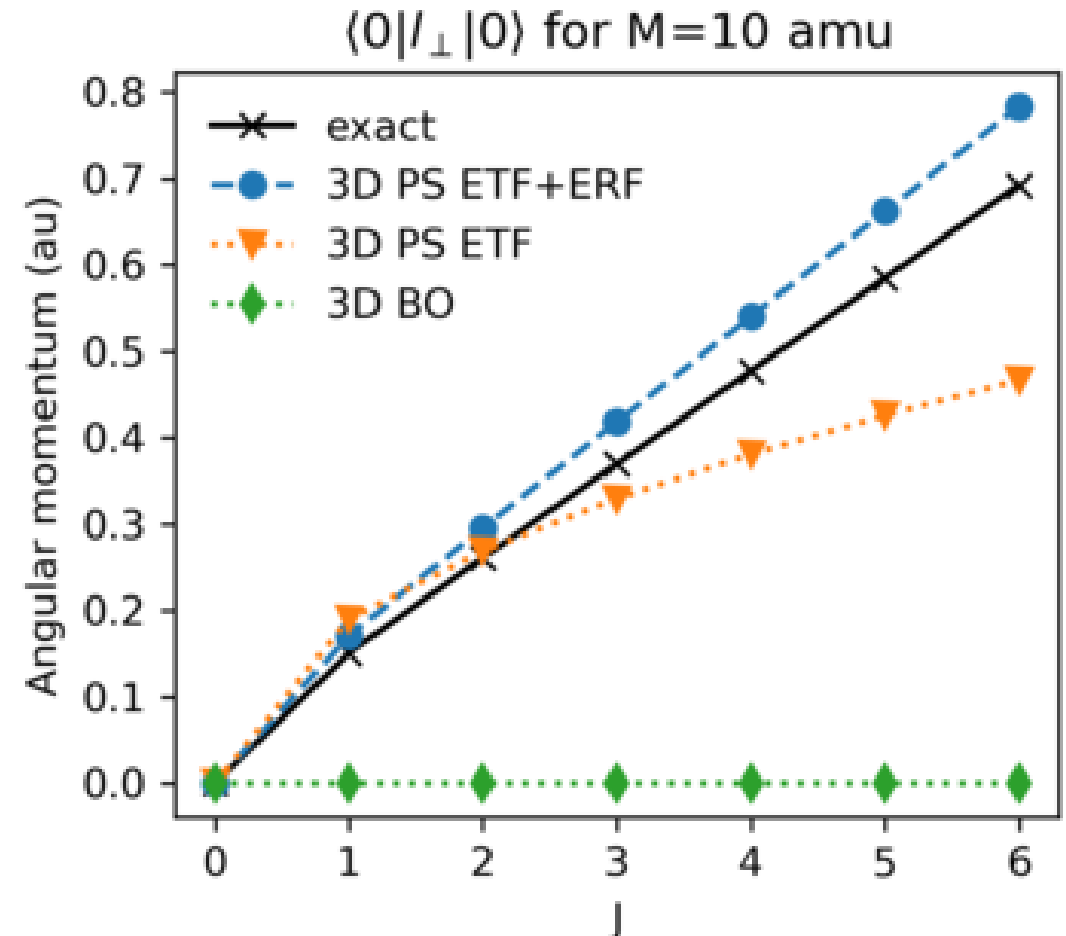


Results: Electronic Angular Momentum for different J values

ETF : Electron Translation Factor

ERF : Electron Rotation Factor

- At $J=0$ all three methods match since $\langle \psi_0 | \hat{l}_e | \psi_0 \rangle = 0$
- At higher J values, molecule is rotating fast so nuclear and electronic momentum get strongly coupled. Thus, non-zero momentum even in the ground state.
- Born-Oppenheimer completely misses it
- We need **ETF and ERF** terms to recover correct angular momentum



Summary

- ❑ **Phase-space electronic structure extends Born–Oppenheimer (BO)** framework by including nuclear position and momentum, revealing electronic momentum which is hidden in standard BO theory
- ❑ This leads to **richer physics**, including improved vibrational spectra, rotational spectra, Raman Optical Activity, Lambda-Doubling, Spin-Rotation Coupling.

Tao, Z., **Bhati, M.** and Subotnik, J.E. *APL Computational Physics* (2026)

Peng, L. et al. (incl. Bhati, M.), *J. Phys. Chem. Lett.* (2026)

Peng, L. et al. (incl. Bhati, M.), *arXiv preprint*, 2603.13211 (2026) (Submitted)

- ❑ Because total angular momentum is naturally conserved, phase-space methods enable momentum transfer between nuclei, electrons, and spin - opening new possibilities for studying **chiral phenomena**.
- ❑ The proposed **Γ operator** can be added into existing electronic-structure theories at nearly the same cost as BO, allowing **electron-nuclear correlation** to be captured both far and near conical intersections.

Future Directions

Understand Non-Adiabatic Coupling spin terms

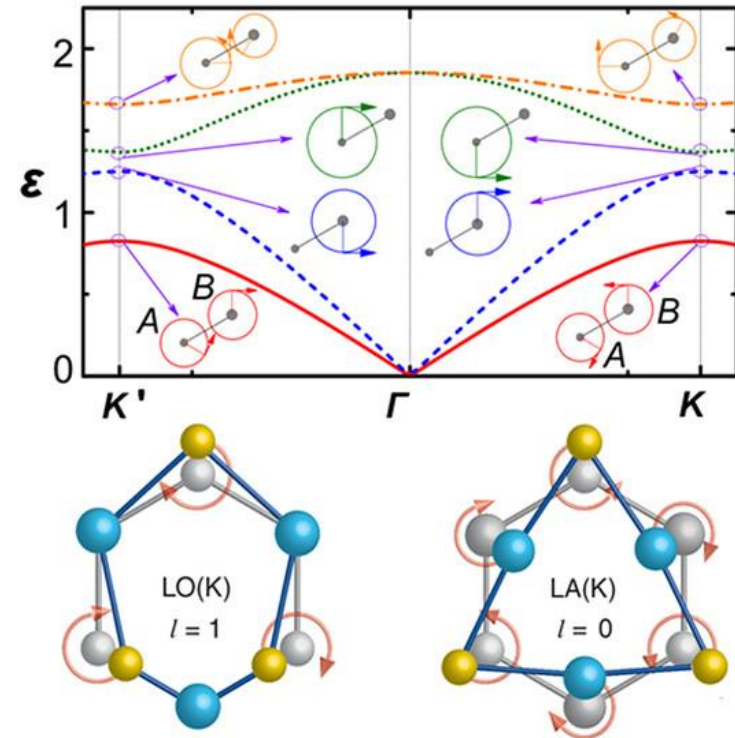
$$\hat{H}_{rot} = \frac{1}{2R^2} (\hat{\mathbf{J}} - \hat{\mathbf{l}} - \hat{\mathbf{s}})^2$$

$\hat{\mathbf{l}} \cdot \hat{\mathbf{s}}$ Spin-orbit coupling

$\hat{\mathbf{J}} \cdot \hat{\mathbf{s}}$ Spin-rotation coupling

$\hat{\mathbf{J}} \cdot \hat{\mathbf{l}}$ Coriolis coupling

Extend to Periodic Systems



Wang, T., Sun, H., Li, X. and Zhang, L., 2024. *Nano Letters*, 24(15), pp.4311-4318.

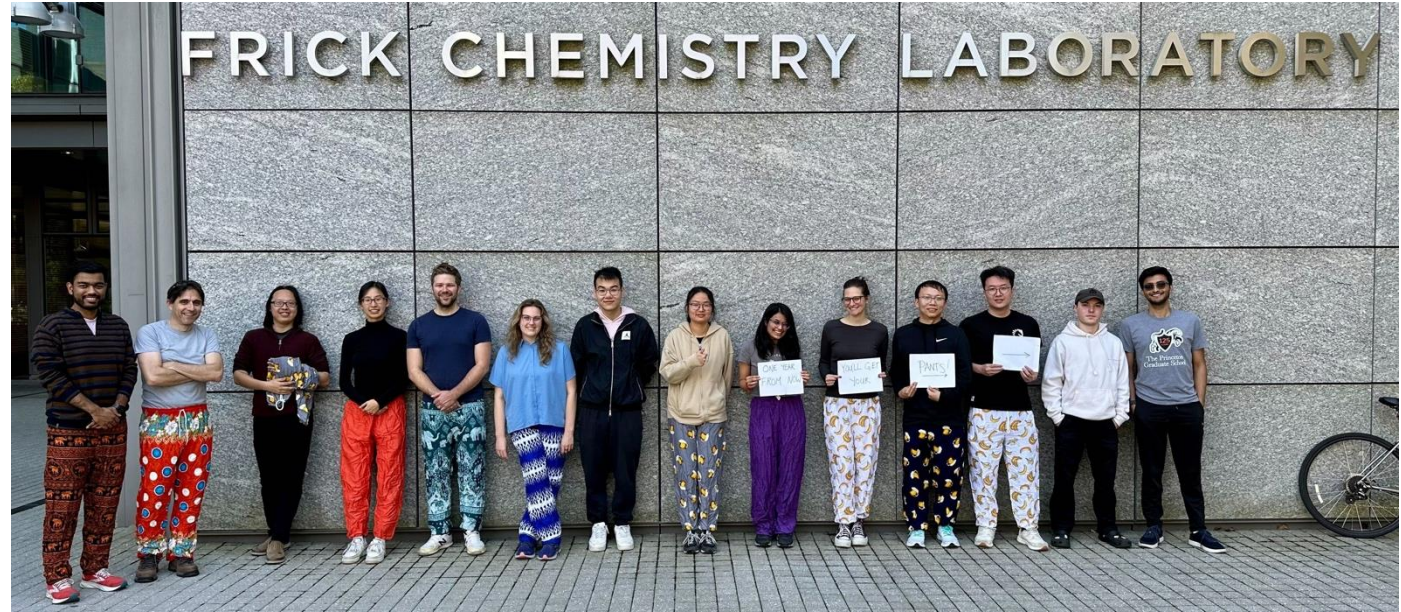
Acknowledgement



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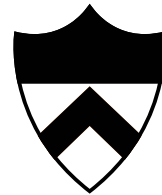
Dr. Nadine Bradbury



Dr. Vale Cofer-Shabica



Prof. Robert Littlejohn
(UC Berkeley)

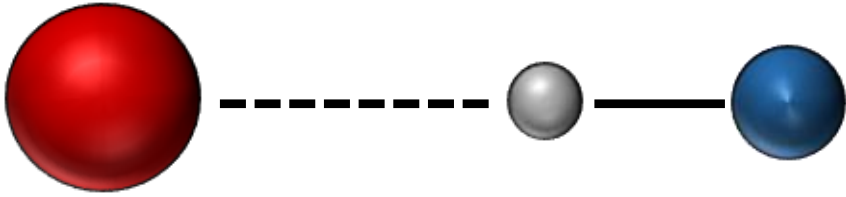


**RESEARCH
COMPUTING**

Taylor fellowship

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Results: Comparing Nuclear Momentum



Electron Translation Factor (ETF)

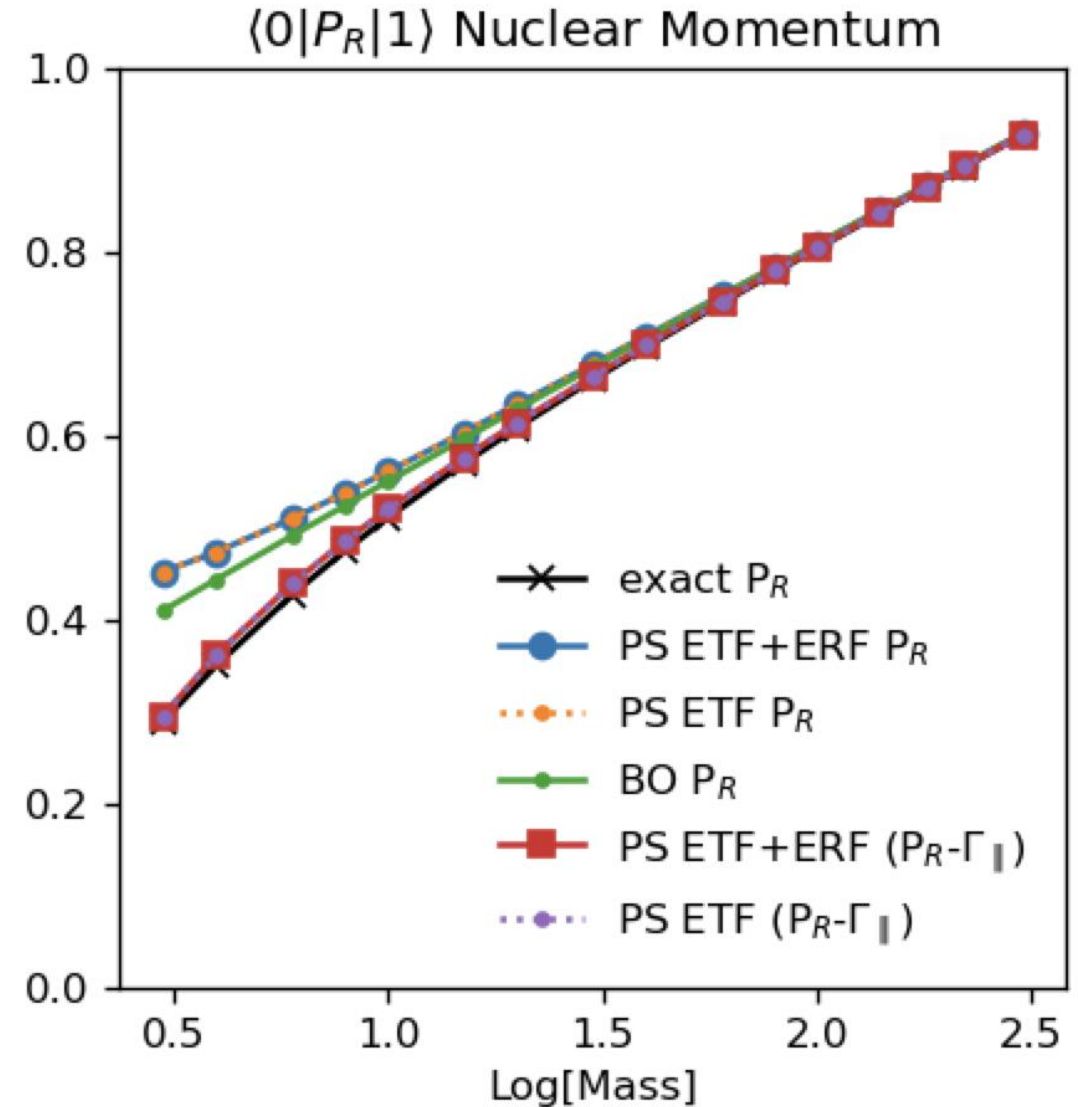
$$\hat{\Gamma}'_A = \frac{1}{2i\hbar} \left(\hat{\Theta}_A \hat{\mathbf{p}} + \hat{\mathbf{p}} \hat{\Theta}_A \right)$$

Corrects for total linear momentum

Electron Rotational Factor (ERF)

$$\hat{\Gamma}''_A = - \sum_B \zeta_{AB} (\mathbf{X}_A - \mathbf{X}_B^0) \times \left(\mathbf{K}_B^{-1} \frac{1}{2i\hbar} \left(\hat{\Theta}_B \hat{\mathbf{l}}_{eB} + \hat{\mathbf{l}}_{eB} \hat{\Theta}_B \right) \right)$$

Corrects for total angular momentum

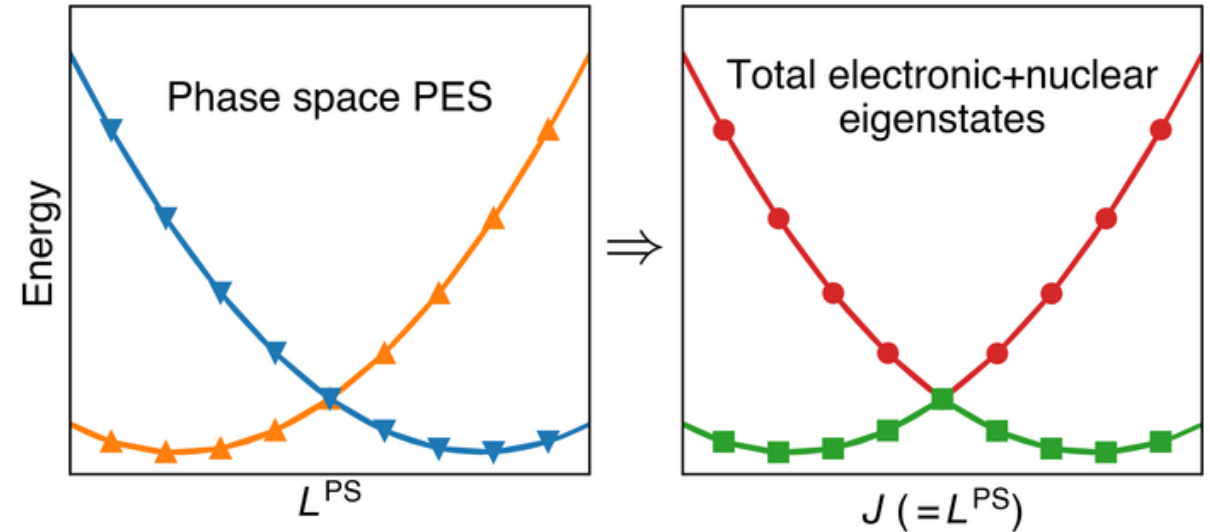


Comparing Phase-Space with Exact approach

Total Angular momentum is conserved

$$J_{PS} = L_{nx}^2 + L_{ny}^2 + L_{nz}^2 = P_{\theta}^2 + \frac{P_{\phi}^2}{\sin^2 \theta_R}$$

$$J(J+1) = P_{\theta}^2 + \frac{P_{\phi}^2}{\sin^2 \theta_R}$$



PS Nuclear Momentum = Total electronic+nuclear momentum

Weyl Transform

$$\left\langle R \left| \hat{H}_{\text{vib}}^{PS} \right| R' \right\rangle = \int \frac{d\mathbf{P}_R}{2\pi\hbar} e^{\frac{i}{\hbar} \mathbf{P} \cdot (\mathbf{R} - \mathbf{R}')} E_{PS} \left(\frac{\mathbf{R} + \mathbf{R}'}{2}, \mathbf{P}_R \right)$$